

Community Help Board – UML Design

1. Use Case Diagram (Text Representation)

Actors:
- Guest
- User (Logged in)

Use Cases:
Guest:
- View Help Requests

User:
- Register / Login
- Create Help Request
- Reply to Help Request
- Mark Reply as Helpful
- Mark Help Request as Solved

2. Class Diagram

```
User
-----
id
name
email
password
reputation
createdAt

HelpRequest
-----
id
title
description
category
status (open / solved)
createdAt
userId (FK)

Reply
-----
id
content
isHelpful
createdAt
userId (FK)
helpRequestId (FK)

Relationships:
- User 1..* HelpRequest
- User 1..* Reply
- HelpRequest 1..* Reply
```

3. Sequence Diagram (Post Help Request)

```
User -> Frontend : Write help request
Frontend -> Backend : POST /requests
Backend -> Backend : Validate JWT
Backend -> Database : Save HelpRequest
Database -> Backend : OK
Backend -> Frontend : Response
Frontend -> User : Show new request
```